5 Essential Authentication Features - Cheat Sheet

1⃣ Overview

Authentication is more than just logging in; it involves multiple security layers, user experience considerations, and compliance requirements. Here are five crucial authentication features every modern application should implement:

2⃣ Key Authentication Features

✅ 1. Login & Registration

- Secure credential storage with **password hashing (BCrypt, Argon2, PBKDF2)**
- Customizable user flows for seamless onboarding
- Protection against **brute-force attacks** and **credential stuffing**

✅ 2. Single Sign-On (SSO)

- Allows users to log in once and access multiple applications
- Uses **OAuth 2.0, OpenID Connect (OIDC), and SAML**
- Reduces password fatigue and enhances security

✅ 3. Multi-Factor Authentication (MFA)

- Adds an extra layer of security beyond passwords
- Methods include **TOTP (Google Authenticator), biometrics, push notifications**
- Prevents unauthorized access even if passwords are compromised

✅ 4. Passkeys (WebAuthn)

- Enables passwordless authentication using **biometrics and device-native security**
- Eliminates phishing risks associated with passwords
- Works with modern browsers and platforms for a seamless login experience

✅ 5. Magic Links

- One-time login links sent via **email**
- No need for passwords; enhances **user experience**
- Reduces password-related security risks

3⃣ Implementation Considerations

🔧 **Challenges of Building from Scratch:**

- Requires expertise in **security best practices, UI/UX, email systems, and compliance**
- Time-intensive and prone to security vulnerabilities

🛠️ **Solution: Authentication Providers (CIAM Solutions)**

- Providers like **FusionAuth, Auth0, Okta** abstract away complexity
- Offer **built-in security, customization, and compliance features**

4️⃣ Learning Resources

📺 **YouTube:**

- OAuth 2.0 and OpenID Connect Explained
- Multi-Factor Authentication Explained
- WebAuthn & Passkeys

🎓 **Pluralsight:**

- Identity & Access Management Fundamentals
- Building Secure Authentication Systems

📝 **Company Blogs:**

- FusionAuth Blog
- Auth0 Blog
- Okta Developer Blog
- WebAuthn Overview

🚀 **Summary:**

- **Use Login & Registration** for secure onboarding
- **Implement SSO** to streamline authentication across apps
- **Enable MFA** to strengthen security
- **Adopt Passkeys (WebAuthn)** for a passwordless experience
- **Leverage Magic Links** for frictionless authentication

💡 **Need a CIAM solution?** Consider **FusionAuth** for flexible deployment, comprehensive security, and a generous free tier! 🔥

To generate a cheat sheet for any topic, follow these steps:

1⃣ Structure the Cheat Sheet

Break down the topic into key sections. A good structure includes:

- **Overview**: A brief explanation of the concept.
- **Key Features**: List the main points with concise descriptions.
- **Implementation Considerations**: Highlight best practices and challenges.
- **Learning Resources**: Provide links to articles, videos, and company blogs.
- **Summary**: A quick recap of the most important points.

2⃣ Use Clear Formatting

- Use **bold headings** for readability.
- Include **bullet points** for quick scanning.
- Provide **links to external resources** for deeper learning.

3⃣ Tools to Generate a Cheat Sheet

- **Manually**: Write it in a document or use Markdown for structuring.
- **AI Assistance**: Use ChatGPT to summarize and format key points.
- **Online Tools**:
    - Cheatography – Premade and customizable cheat sheets.
    - Markdown Editors – Create structured cheat sheets.

Would you like me to generate another cheat sheet for a specific topic? 🚀